Table 5. Random parameters binary logit model with heterogeneity in means and variances of crash injury severity (serious injury/fatal vs. PDO/minor injury), crashes involving a motorcycle and another vehicle type, Thailand, 2016–2023 (yearly periods), adaptive fused temporal structure.

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Constant [serious injury / fatal]				
Constant [2016–2019]	0.225	1.31		
Constant [2020–2023]	–0.056	–0.34		
Variables producing the same parameter value across all periods [2016–2023]				
Vehicle involvement				
One motorcycle involved (1 if yes, 0 otherwise) [2016–2023]	–0.516	–3.90	0.0633	–0.0633
One pickup truck involved (1 if yes, 0 otherwise) [2016–2023]	–0.048	–0.55	–0.0125	0.0125
(Standard deviation of parameter distribution; 48.6% above zero)	1.374	8.45		
Two or more pickup trucks involved (1 if yes, 0 otherwise) [2016–2023]	–0.482	–3.19	0.0574	–0.0574
One 6-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	1.081	8.64	–0.1338	0.1338
One 10-wheel truck involved (1 if yes, 0 otherwise) [2016–2023]	1.448	10.43	–0.1786	0.1786
One truck with more than 10 wheels involved (1 if yes, 0 otherwise) [2016–2023]	3.281	7.08	–0.2567	0.2567
(Standard deviation of parameter distribution; 79.9% above zero)	3.910	5.97		
Crash type				
Sideswipe (parallel movement) (1 if yes, 0 otherwise) [2016–2023]	–0.853	–11.38	0.1026	–0.1026
Head-on collision (1 if yes, 0 otherwise) [2016–2023]	1.298	10.68	–0.1617	0.1617
Crash cause				
Cutting in too closely (1 if yes, 0 otherwise) [2016–2023]	0.032	0.45	–0.0110	0.0110
(Standard deviation of parameter distribution; 51.0% above zero)	1.256	6.53		
Running a red light (1 if yes, 0 otherwise) [2016–2023]	0.782	5.49	–0.0962	0.0962
Roadway characteristics				
Raised or earth median (1 if yes, 0 otherwise) [2016–2023]	–0.417	–6.79	0.0505	–0.0505
Depressed median (1 if yes, 0 otherwise) [2016–2023]	0.459	5.82	–0.0550	0.0550
(Standard deviation of parameter distribution; 63.0% above zero)	1.386	6.25		
Barrier median (1 if yes, 0 otherwise) [2016–2023]	–1.076	–12.26	0.1297	–0.1297
Horizontal curve (1 if yes, 0 otherwise) [2016–2023]	0.357	3.92	–0.0438	0.0438
Concrete pavement surface (1 if yes, 0 otherwise) [2016–2023]	–0.465	–6.65	0.0564	–0.0564
T-intersection (1 if yes, 0 otherwise) [2016–2023]	–0.099	–0.84	0.0008	–0.0008
(Standard deviation of parameter distribution; 48.0% above zero)	1.949	5.13		
U-turn without auxiliary lane (1 if yes, 0 otherwise) [2016–2023]	0.356	2.35	–0.0436	0.0436
U-turn with deceleration lane only (1 if yes, 0 otherwise) [2016–2023]	0.583	4.06	–0.0715	0.0715
U-turn with deceleration and acceleration lanes (1 if yes, 0 otherwise) [2016–2023]	0.516	3.04	–0.0633	0.0633
Environmental characteristics				
Dark, no street lighting (1 if yes, 0 otherwise) [2016–2023]	1.007	9.93	–0.1254	0.1254
Temporal characteristics				
Songkran festival period (1 if yes, 0 otherwise) [2016–2023]	–0.801	–5.31	0.0405	–0.0405
(Standard deviation of parameter distribution; 42.3% above zero)	4.140	7.62		
Midnight / early morning (1 if yes, 0 otherwise) [2016–2023]	0.741	8.46	–0.0911	0.0911
Region				
Eastern region (1 if yes, 0 otherwise) [2016–2023]	1.393	13.67	–0.1718	0.1718
Northeastern region (1 if yes, 0 otherwise) [2016–2023]	1.523	19.16	–0.1939	0.1939
Western region (1 if yes, 0 otherwise) [2016–2023]	1.137	10.68	–0.1404	0.1404
Variables with temporally varying parameters				
Vehicle involvement				
One passenger car involved (1 if yes, 0 otherwise) [2016–2020]	–0.438	–5.19	0.0535	–0.0535
One passenger car involved (1 if yes, 0 otherwise) [2021–2023]	–0.621	–6.51	0.0786	–0.0786
Crash type				
Rear-end collision in the same lane (1 if yes, 0 otherwise) [2016–2018]	–0.397	–4.30	0.0452	–0.0452
Rear-end collision in the same lane (1 if yes, 0 otherwise) [2019–2023]	–0.129	–1.75	0.0162	–0.0162
Crash cause				

Variable description	Estimated parameter	*t*-statistic	Marginal effects	
			PDO / minor injury	Serious injury / fatal
Speeding (1 if yes, 0 otherwise) [2016–2018]	–0.269	–3.06	0.0308	–0.0308
Roadway characteristics				
Crash on a frontage road (1 if yes, 0 otherwise) [2016–2017]	–2.124	–7.86	0.2174	–0.2174
Crash on a frontage road (1 if yes, 0 otherwise) [2018–2023]	–1.102	–8.14	0.1326	–0.1326
Environmental characteristics				
Raining (1 if yes, 0 otherwise) [2016–2017]	1.318	5.02	–0.1517	0.1517
Raining (1 if yes, 0 otherwise) [2018–2023]	0.219	1.66	–0.0273	0.0273
Dark, with street lighting (1 if yes, 0 otherwise) [2016–2019]	0.148	1.75	–0.0171	0.0171
Dark, with street lighting (1 if yes, 0 otherwise) [2020–2023]	0.373	4.59	–0.0473	0.0473
Temporal characteristics				
New Year festival period (1 if yes, 0 otherwise) [2016–2018]	–0.792	–3.18	0.0227	–0.0227
(Standard deviation of parameter distribution; 44.4% above zero)	5.574	5.05		
New Year festival period (1 if yes, 0 otherwise) [2019–2023]	–1.191	–6.84	0.0874	–0.0874
(Standard deviation of parameter distribution; 36.3% above zero)	3.398	7.02		
Region				
Northern region (1 if yes, 0 otherwise) [2016–2019]	0.657	5.45	–0.0769	0.0769
Northern region (1 if yes, 0 otherwise) [2020–2023]	–0.320	–3.49	0.0403	–0.0403
Southern region (1 if yes, 0 otherwise) [2016–2017]	0.236	1.75	–0.0265	0.0265
Southern region (1 if yes, 0 otherwise) [2018–2023]	1.018	11.38	–0.1287	0.1287
Heterogeneity in the means of random parameters				
One pickup truck involved [2016–2023]: Evening [2016–2023]	0.376	3.48		
Depressed median [2016–2023]: Wet road surface [2018–2023]	–0.783	–2.64		
Cutting in too closely [2016–2023]: Van involved [2016–2023]	0.721	2.10		
Heterogeneity in the variances of random parameters				
One truck with more than 10 wheels involved [2016–2023]: Peak hours [2016–2023]	0.450	2.46		
Model statistics				
Number of observations		21,371		
Number of estimated parameters		56		
Log-likelihood at zero, LL(0)		–14813.25		
Log-likelihood with constant only, LL(C)		–14745.16		
Log-likelihood at convergence, LL(β)		–12052.03		
McFadden pseudo $\rho^2 = 1 - LL(\beta)/LL(0)$		0.186		
AIC / BIC		24,216.1 / 24,662.4		
Random parameter distribution; simulation		Normal; 2,000 Halton draws		

Notes. A positive parameter increases the probability of a serious injury/fatal outcome. Reference categories: Central region, daytime, no median. Periods in brackets share one parameter value; for a variable with temporally varying parameters, periods not shown have a parameter of zero. Temporal structure re-selected by the adaptive fused lasso inside the random parameters model: the sparsest penalty whose structure is not rejected at the 90% level against the fully unconstrained (separate period) random parameters models ($LR < \chi^2_{0.90}(df)$); candidate structures were rebuilt and pruned in order of sparsity until one was not rejected (Table D1). The stage at which the structure was chosen (fixed-parameter or random parameters model) is the one with the lower BIC among models not rejected at the 90% level against the fully unconstrained models (Table 3; all candidates in Table D1). Random parameters (at most eight), heterogeneity in means (at most three terms) and in variances (at most one term) were searched with 100 Halton draws at the 90% level and confirmed with 2,000 draws; non-significant parameters were then removed, so all parameters except constants and the means of random parameters are statistically significant at the 90% level (a parameter is kept random only while its standard deviation is significant). A variable used in a heterogeneity function enters only the random parameter shown, for crashes in the shared periods, and is removed from the main utility. Marginal effects are average changes in the simulated unconditional probability (2,000 Halton draws) when the indicator switches from 0 to 1, averaged over crashes in the parameter's periods; for random parameters they include the heterogeneity terms. PDO/minor = –(serious/fatal). Percentage above zero evaluated at the heterogeneity variables equal to zero.